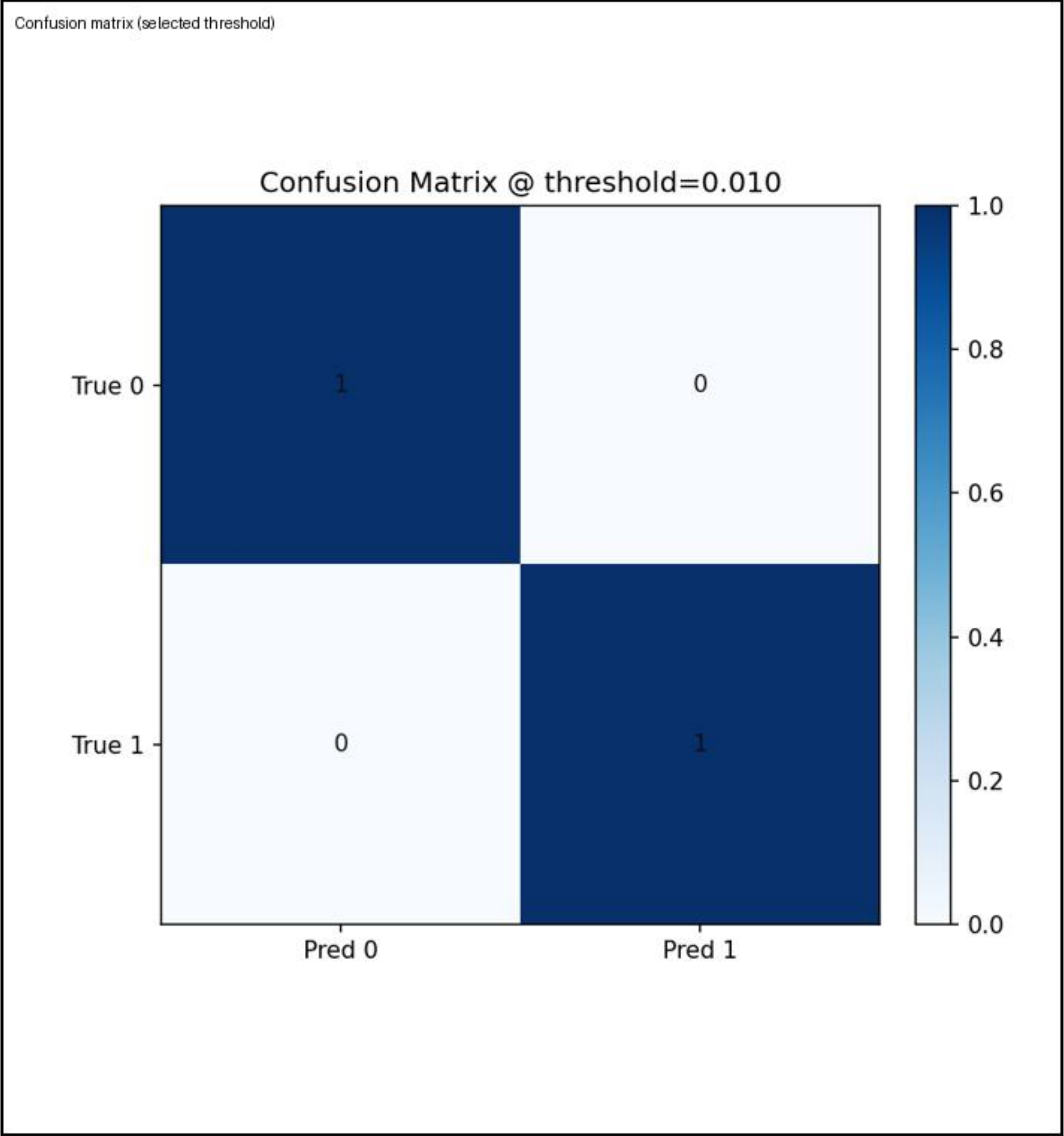
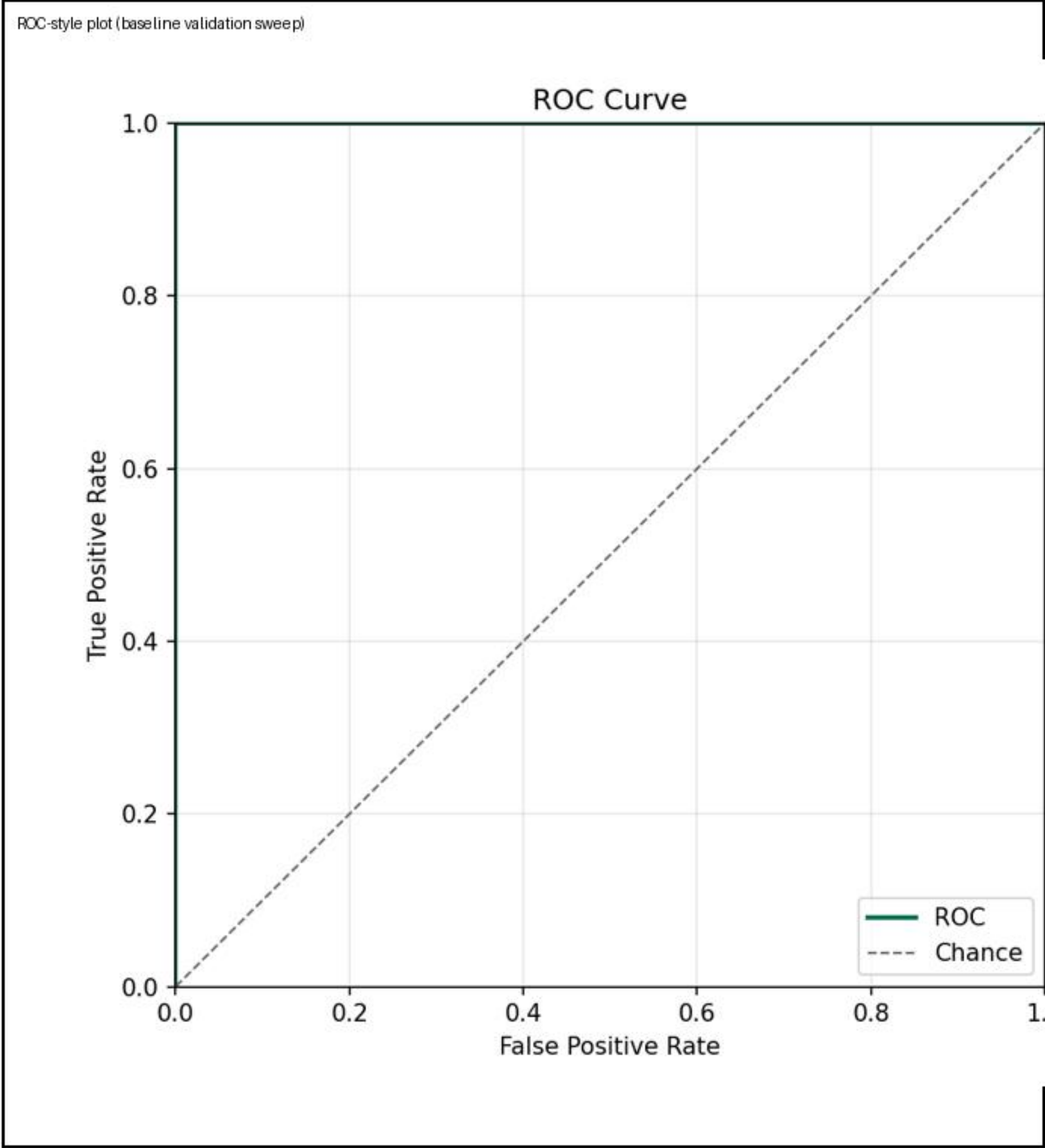


Milestone 2 Evaluation Report

Overview: deterministic face-verification evaluation with tracked runs, validation-based threshold selection, error analysis, and one data-centric change (prefer_unique positive pair sampling).

Threshold rule: maximize balanced accuracy on the validation split, then lock that threshold for test reporting.

Tracked evidence: five runs are summarized in reports/run_comparison.md and copied as lightweight JSON artifacts in reports/tracked_runs/.



Baseline versus improved summary: the baseline and improved configs both reach balanced accuracy 1.00 on the separable synthetic runs, but the improved config changes the pair composition rather than the evaluation code. That makes the comparison traceable and keeps the threshold-selection rule fair across versions. The imperfect runs are retained to expose concrete failure modes for slice analysis.

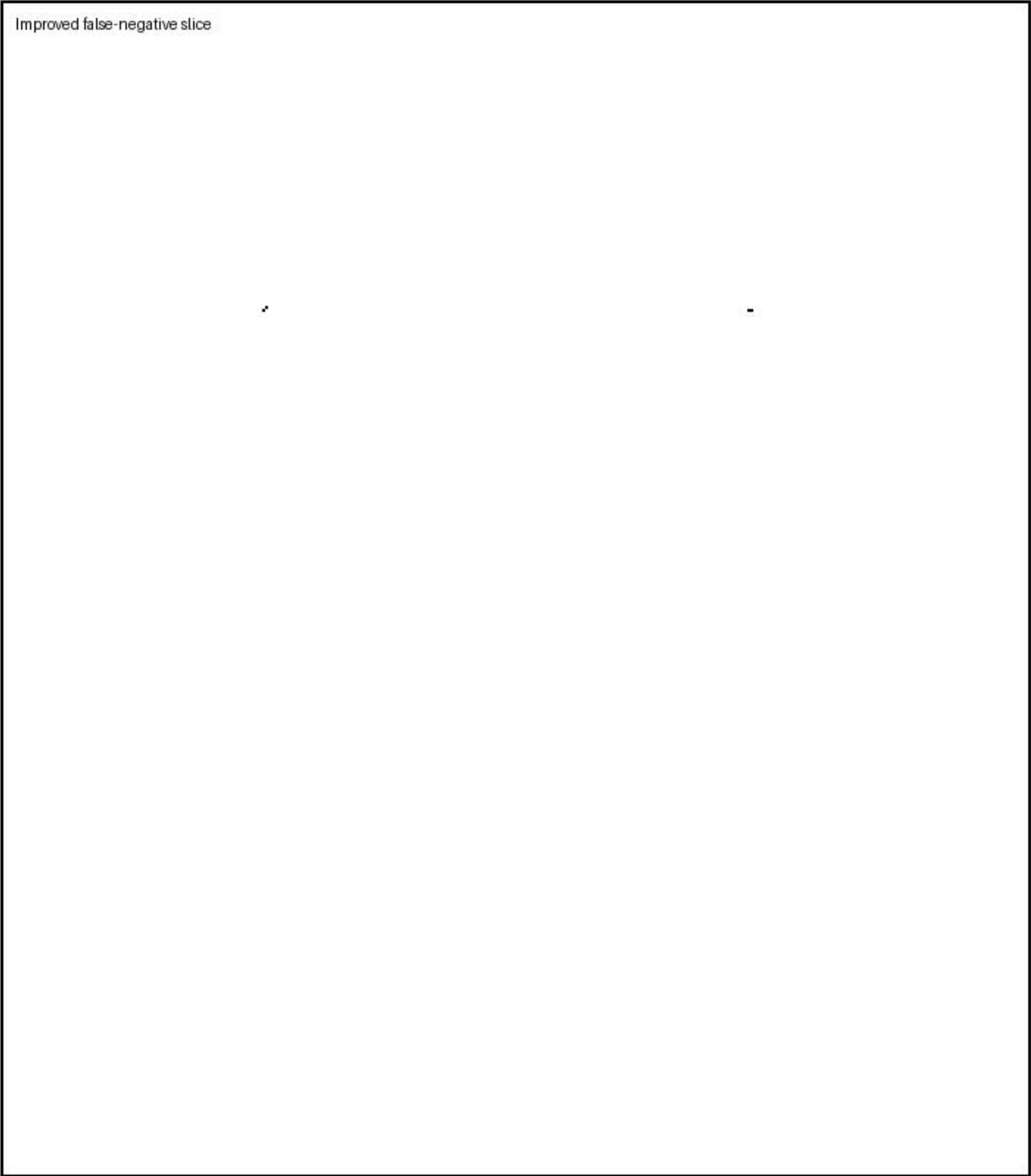
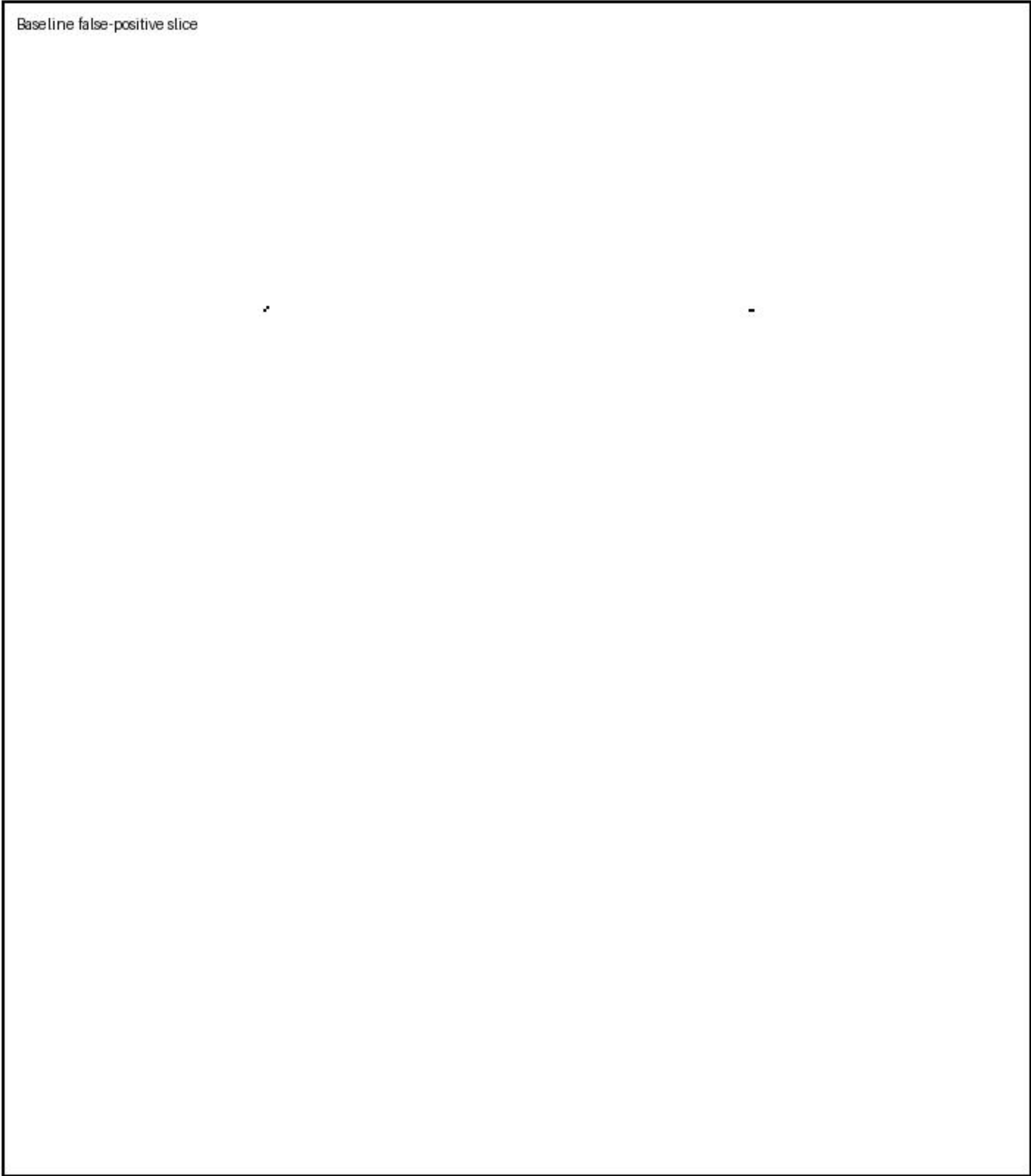
Data-centric change: configs/m2_improved.yaml sets positive_pair_strategy=prefer_unique. This reduces repeated positive-pair reuse and better distributes same-identity evidence across unique image combinations.

Submission evidence: reports/milestone2_report.pdf, reports/run_comparison.csv, reports/tracked_runs_summary.json, and reports/tracked_runs/<run_id>/.

Error Slice Analysis

Slice 1: high-similarity false positives. Definition: different-identity pairs scoring above the selected threshold. Count: 1 example in the baseline error-analysis run. Hypothesis: grayscale flatten-and-cosine features overemphasize coarse shared structure and under-model identity detail. Future improvement: stronger embeddings or harder negative sampling.

Slice 2: low-similarity false negatives. Definition: same-identity pairs scoring below the selected threshold. Count: 1 example in the improved error-analysis run. Hypothesis: the baseline representation is brittle to variation analogous to pose, lighting, or expression changes. Future improvement: replace raw pixel features while keeping the improved pair policy.



Counts and examples are also recorded in `reports/error_slices/baseline_error/slices.json` and `reports/error_slices/improved_error/slices.json`. These slice directories provide non-cherry-picked evidence tied back to tracked runs.

Main lesson: Milestone 2 turned the prototype into a small reproducible ML system. The threshold is selected deliberately, tracked runs are recorded with code/data versions, and the data-centric change is evaluated under the same rule rather than through informal experimentation.